

Lab 1: Data and statistics

In this lab, we will begin by working with data to calculate fundamental statistics on a real-life dataset. You will develop a data loader to import the dataset. Utilizing functions you have created, you will gain valuable insights from the dataset and generate data plots for visualization purposes.

The data to be used is available in Canvas and shows information about different housing districts. The name of the file is **housing.csv**.

Task 1: Load and inspect the housing data

- Load the housing data contained as a resource provided for this lab: **housing.csv**. You can find it on canvas or download it from [link](#).
 - You can use built-in functions

```
import pandas as pd

pd.read_csv("housing.csv")
```

- The output of the load function should look like this:

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY
5	-122.25	37.85	52.0	919.0	213.0	413.0	193.0	4.0368	269700.0	NEAR BAY
6	-122.25	37.84	52.0	2535.0	489.0	1094.0	514.0	3.6591	299200.0	NEAR BAY
7	-122.25	37.84	52.0	3104.0	687.0	1157.0	647.0	3.12	241400.0	NEAR BAY
8	-122.26	37.84	42.0	2555.0	665.0	1206.0	595.0	2.0804	226700.0	NEAR BAY
9	-122.25	37.84	52.0	3549.0	707.0	1551.0	714.0	3.6912	261100.0	NEAR BAY
10	-122.26	37.85	52.0	2202.0	434.0	910.0	402.0	3.2031	281500.0	NEAR BAY

This dataset provides a set of housing districts and their information. Take a moment to understand what information it shows.

Task 2: Explore the dataset to answer the research questions

Develop your own functions to explore the housing dataset to answer the questions below. Always take a moment to interpret each question, and when you have the results, always think about their meaning. When presenting your solution, focus both on implementation and the interpretation of the results.

1. How many geographical units are there in the dataset?
2. What is the mean house value among all "Ocean_proximity" categories?
3. What is the mean house value in each "Ocean_proximity" category?
4. What can you tell about the differences between the mean and median values "Ocean_proximity" categories?
5. Create a histogram for "households", "median_income", "housing_median_age" and "median_house_value".
 - An example of how to create a histogram using matplotlib is shown here:

```
from matplotlib import pyplot as plt
```

```
grades = [8, 6, 1, 7, 8, 9, 8, 7, 10, 7, 6, 9, 7]
```

```
plt.hist(grades)
```

```
plt.title("histogram")
```

```
plt.show()
```

6. What do you notice about the graphs? Specifically focus on end of "housing_median_age" and "median_house_value" graphs". What do you think about the magnitude of the values in "median_house_value"?
7. In which category are the most expensive and the largest houses typically located? In which category are the least expensive and smallest houses typically located?
8. What can you tell about the quality of houses in each "Ocean_proximity" category? (e.g., the age of the houses, the number of rooms)
9. What can you tell about the demographic living in each "Ocean_proximity" category? (e.g., population, households, median income)
10. Are there any other interesting observations and questions you can come up with regarding this dataset?